

Exam 1 Practice

MATH 2025

Answer the questions in the spaces provided. Show your work wherever practicable. You are permitted to use a calculator but not any other resources.

The data set for this exam is a subset from the Spotify Songs Tidy Tuesday data set. The data were originally obtained from Spotify using the **spotifyr** R package. **It is the same data set we used for our activities.**

It contains numerous characteristics for each song. This analysis will focus specifically on the following variables:

variable	class	description
track_popularity	double	Song Popularity (0-100) where higher is better
energy	double	Energy is a measure from 0.0 to 1.0 and represents a perceptual measure of intensity and activity. Typically, energetic tracks feel fast, loud, and noisy. For example, death metal has high energy, while a Bach prelude scores low on the scale. Perceptual features contributing to this attribute include dynamic range, perceived loudness, timbre, onset rate, and general entropy.
valence	double	A measure from 0.0 to 1.0 describing the musical positiveness conveyed by a track. Tracks with high valence sound more positive (e.g. happy, cheerful, euphoric), while tracks with low valence sound more negative (e.g. sad, depressed, angry).
danceability	double	Danceability describes how suitable a track is for dancing based on a combination of musical elements including tempo, rhythm stability, beat strength, and overall regularity. A value of 0.0 is least danceable and 1.0 is most danceable.
playlist_genre	character	The genre of the song (edm, latin, pop, r&b, rap, rock)

```
spotify |>
  select(track_popularity, energy, valence, playlist_genre) |>
  head() |>
  kable()
```

track_popularity	energy	valence	playlist_genre
80	0.859	0.520	pop
80	0.694	0.216	pop
83	0.777	0.706	latin
83	0.499	0.511	rap
89	0.892	0.478	pop
83	0.880	0.534	pop

For questions 1-7, consider `model1` below:

```
model1 <- lm(danceability ~ valence + energy, data=spotify)
model1 |> tidy() |> kable(digits = 3)
```

term	estimate	std.error	statistic	p.value
(Intercept)	0.643	0.022	29.694	0.000
valence	0.260	0.027	9.693	0.000
energy	-0.125	0.037	-3.410	0.001

1. (5 points) Use the values above to write down the equation that represents the regression model. Use the variable names to represent the variables rather than Y 's and X 's.
2. (10 points) Give a one sentence interpretation (in context) of the first two numbers in the estimate column (i.e. 0.643 and 0.260).

3. (5 points) Which of the following represents the predicted **danceability** for a single observation that has a **valence** of 0.5 and an **energy** of 0.25?
- A. $0.643 + 0.260 - 0.125 = 0.778$
 - B. $0.643 + 0.260 \times 0.5 - 0.125 \times 0.25 = 0.7418$
 - C. $29.694 + 9.693 \times 0.25 - 3.410 \times 0.5 = 30.4123$
 - D. $0.260 \times 0.25 - 0.125 \times 0.5 = 0.0025$
 - E. Not enough information
4. (5 points) What is the residual of this estimated **danceability**?
- A. -0.240
 - B. 0.143
 - C. 0.375
 - D. Not enough information to say
5. (5 points) Suppose you generate a 95% prediction interval for this prediction. If we change this prediction interval to a confidence interval (that is make a prediction for the average of your response rather than a single prediction) the width of the interval would:
- A. increase
 - B. decrease
 - C. remain unchanged (approximately)
6. (15 points) Suppose you use bootstrapping to generate a 95% confidence interval for the slope of **valence** in **model1**. For each of the following changes, indicate whether the width of your interval would increase, decrease, or approximately remain the same (keeping all else constant):
- (a) Increase sample size:
 - A. increase
 - B. decrease
 - C. remain unchanged (approximately)
 - (b) Increase confidence level:
 - A. increase
 - B. decrease
 - C. remain unchanged (approximately)
 - (c) Increase the number of bootstrapped replicates:
 - A. increase
 - B. decrease
 - C. remain unchanged (approximately)

7. (15 points) Consider a hypothesis test to determine whether there is a relationship between `valence` and `danceability` in `model1`:

(a) State the null and alternative hypothesis in both words and symbols.

(b) Give a rough definition of a p-value in the context of this problem.

(c) What would be the result of the test in the context of the problem if $\alpha = 0.05$?

For question 8, consider `model2` below:

```
model2 <- lm(danceability ~ valence + energy + playlist_genre, data=spotify)
```

8. (5 points) Recall that `playlist_genre` has 6 distinct genre's. How many *additional* parameters (i.e. β 's) would `model2` include that `model1` would not?

- A. 5
- B. 6
- C. 10
- D. 12
- E. 15
- F. 18

9. (10 points) For the constant variance assumption of a linear model, sketch a plot that would show that this assumption would be violated. Make sure to clearly label your axes so I know what the plot represents. Feel free to also describe what I should be looking for in words if you feel that your plot is not sufficiently clear.

The code below will be helpful for the rest of the exam.

```
spotify <- spotify |>
  mutate(pop_indicator = as.factor(ifelse(playlist_genre == "pop", "pop", "not pop")))
```

10. (8 points) Consider the following model:

```
model1 <- lm(danceability ~ energy*pop_indicator, data=spotify)

model1 |> tidy() |> kable()
```

term	estimate	std.error	statistic	p.value
(Intercept)	0.6478241	0.0312549	20.727138	0.0000000
energy	0.1053857	0.0489308	2.153772	0.0317316
pop_indicatorpop	0.0477146	0.0464676	1.026837	0.3049900
energy:pop_indicatorpop	-0.1266868	0.0710332	-1.783487	0.0751086

This model contains two different lines. One for **pop** songs and one for **not pop**. Write down the equations of both lines. For partial credit you may just write down the equation of the model.

(a) **pop** line:

(b) **not pop** line:

For questions 11-18, consider the following models:

```
model1 <- lm(danceability ~ energy*pop_indicator,  
             data=spotify)  
model2 <- lm(danceability ~ valence, data=spotify)  
model3 <- lm(danceability ~ valence + energy + pop_indicator, data=spotify)  
model4 <- lm(danceability ~ valence + track_popularity + energy*pop_indicator, data=spotify)
```

11. (5 points) Which of the following metrics is the LEAST useful for comparing these four models:
- A. R^2
 - B. R^2_{adj}
 - C. AIC
 - D. BIC
 - E. C_p
12. (5 points) For the answer you selected above, which of the following is true (select all that apply):
- A. It is fine to use for comparing models with the same number of parameters.
 - B. Using it can lead to overfitting because increasing the model complexity will always lead to a “better” number.
 - C. We only use it when we are looking at a model with one predictor.
 - D. Even though it isn’t useful for comparing models of different sizes, it is useful for describing a model’s quality once a model is chosen.
 - E. It penalizes models for being more complex.
13. (6 points) Which of the following models are nested in one another (select all that apply):
- A. model1 in model2
 - B. model1 in model3
 - C. model1 in model4
 - D. model2 in model3
 - E. model2 in model4
 - F. model3 in model4

For questions 14 - 18 consider the following output in addition to the code at the beginning of the previous page:

Model	r.squared	adj.r.squared	AIC	BIC
model1	0.023	0.017	-623.009	-601.857
model2	0.139	0.137	-691.201	-678.509
model3	0.169	0.164	-705.188	-684.035
model4	0.174	0.166	-704.189	-674.576

14. (2 points) Which model is “best” according to R^2 ?
- A. model1
 - B. model2
 - C. model3
 - D. model4
15. (2 points) Which model is “best” according to R_{adj}^2 ?
- A. model1
 - B. model2
 - C. model3
 - D. model4
16. (2 points) Which model is “best” according to AIC?
- A. model1
 - B. model2
 - C. model3
 - D. model4
17. (2 points) Which model is “best” according to BIC?
- A. model1
 - B. model2
 - C. model3
 - D. model4
18. (3 points) Which model do you think is “best”? If there are two candidate models select the one which is more “parsimonious”? Very briefly state your rationale.
- A. model1
 - B. model2
 - C. model3
 - D. model4